

The giant-component conjecture for the hexagon-flip graph of pyrochlore ice

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Abstract

The local Markov chain on zero-flux pyrochlore ice states, which reverses the six arrows of a hexagon whenever they run head to tail, has a disconnected state space: frozen states, with no flippable hexagon, exist at every cell size and are isolated vertices of the flip graph. This note states and tests the conjecture that they are the only obstruction in cubic geometry, so that in every cubic cell the zero-flux states with at least one flippable hexagon form a single component. The five enumerable cells are censused exactly. At $(1, 2, 2)$, the only one with a two-dimensional cross-section, 221,628 zero-flux states fall into 205 components, a giant of 221,424 and 204 singletons, every singleton frozen, which makes the conjecture exhaustively true there; the quasi-one-dimensional cells shatter instead, so the restriction to thick geometry is doing real work. Beyond enumeration the evidence is 27,000 random zero-flux starts at cubic $L = 2$ and $L = 3$, half fresh and half manufactured next to the frozen set by loop reversals: 13 landed frozen, and every one of the remaining 26,987 was given an explicit path of hexagon flips to the reference state, with none left uncertified. This is a conjecture with computational evidence and no certificate. Nothing here is a proof, and because neither sampler is uniform the evidence does not even bound the density of a hypothetical second non-frozen component.

1 The model and the flip graph

Pyrochlore ice is an arrow field on the bonds of the diamond lattice. Each bond carries one bit, read as an arrow, and the ice rule requires exactly two of the four arrows at each diamond site to point in. The cells used here are $a \times b \times c$ conventional cubic cells with periodic boundaries, carrying $16abc$ bonds, $8abc$ sites and $16abc$ hexagons; the cubic cell of side L is the case $a = b = c = L$, with $16L^3$ spins. Hexagons are derived from adjacency together with displacement rather than taken as any six-cycle, which matters because the smallest cell has 96 six-cycles against 16 true hexagons.

Because the arrow field is divergence-free, the net arrow flux through each of the three coordinate cut planes is conserved by any move that reverses a closed loop. Write $\mathbf{W} = (W_x, W_y, W_z)$ for that flux. All the work below is in the sector $\mathbf{W} = \mathbf{o}$, the largest one.

The chain's only move is the hexagon flip. A hexagon is *flippable* when its six arrows run head to tail around the cycle in one sense or the other, and the move reverses all six; in bit terms the six bond bits alternate around the cycle, and the flip is an exclusive-or with the hexagon's mask. The move preserves the ice rule and $\mathbf{W}$, and it is its own inverse, so it defines an undirected graph. The *flip graph* has the $\mathbf{W} = \mathbf{o}$ ice states of a cell as its vertices and one edge per flippable hexagon per state. A state with no flippable hexagon is *frozen*, and is an isolated vertex. In Rokhsar–Kivelson language a frozen state is an exact zero-energy eigenstate degenerate with the equal-weight superposition over its sector, so its existence is a statement about the ground-space degeneracy and not merely about an algorithm (Rokhsar and Kivelson 1988).

Frozen zero-flux states exist at every cell size. Two constructions give them, periodic tiling of the twelve states of the $(1, 1, 1)$ cell and a Klein-type 4×4 matrix construction, and both are theorems with hand proofs and machine-checked unsatisfiability certificates for their cell instances. So the zero-flux sector is never connected, its spectral gap is exactly zero at every size, and every mixing statement about this chain is a statement about one component.

2 The conjecture

Frozen implies isolated, trivially. The conjecture is the converse at the level of components.

(NF $\Rightarrow$ G). In every cubic cell, every non-frozen zero-flux ice state lies in one and the same component of the flip graph. Equivalently: the subgraph induced on the non-frozen zero-flux states is connected, and the zero-flux flip graph has exactly one more component than the cell has frozen zero-flux states.

The second form is the useful one for testing, because it turns a statement about a giant component into an exact count. At the first cubic cell, $(2, 2, 2)$, the number of frozen zero-flux states is known to be 612, obtained by frozen-constrained SAT enumeration

and independently certified by a proof-logging model counter. Under the conjecture the $(2, 2, 2)$ zero-flux flip graph therefore has exactly 613 components, one of them holding all $\sim 2 \times 10^{11}$ remaining states. Nobody can check that count directly; it is recorded because it is the sharpest form the conjecture takes at a cell whose frozen set is fully known.

Two cautions belong with the statement. The phrase ‘giant component’ has to be read by size rather than by label, since global arrow reversal permutes the components and non-singleton components therefore come in equal-size pairs at most cells. And the restriction to cubic geometry is not decoration: the census below shows the conjecture failing outright on the quasi-one-dimensional cells, so no proof can rest on the ice rule alone.

3 The exact census

Five cells have a zero-flux sector small enough to enumerate with a 64-bit state word. For each, the sector is enumerated by depth-first search with winding pruning (`icemix.enumerate.enumerate_ice_states` with `zero_winding_only`), the flip graph is built as a sparse adjacency matrix with the closure of the state set under every flippable move asserted (`icemix.chain.build_flip_graph`), and components are taken with SciPy (`icemix.chain.components`). The full multiset of component sizes was recomputed from scratch for this note; it agrees with the stage-1 census in every entry, and with the stage-4 recomputation at $(1, 2, 2)$.

cell	spins	compon-		giant	giant frac.	frozen
		W = o	ents			
$(1, 1, 1)$	16	12	12	1	8.3 %	12
$(1, 1, 2)$	32	388	70	160	41.2 %	68
$(1, 1, 3)$	48	15,024	540	3,360	22.4 %	528
$(1, 2, 2)$	64	221,628	205	221,424	99.91 %	204
$(1, 1, 4)$	64	637,988	5,492	43,008	6.7 %	5,412

Writing s^m for m components of size s , the full size multisets are 1^{12} at $(1, 1, 1)$; $1^{68}, 160^2$ at $(1, 1, 2)$; $1^{528}, 432^2, 864^8, 3360^2$ at $(1, 1, 3)$; $1^{204}, 221424^1$ at $(1, 2, 2)$; and $1^{5412}, 832^2, 2496^{66}, 21120^2, 42240^8, 43008^2$ at $(1, 1, 4)$.

Several things can be read off. Singleton components are exactly the frozen states at every cell, with no exceptions in either direction. Non-singleton sizes occur with even multiplicity everywhere except $(1, 2, 2)$, where the giant is a single component that global arrow reversal maps to itself. No component straddles two flux sectors, checked over all 25, 69 and 137 sectors of the three smallest cells.

The cell that carries the conjecture is $(1, 2, 2)$, the only enumerable cell with a two-dimensional cross-section. There the 204 components outside the giant are all singletons and all frozen, so $(NF \Rightarrow G)$ is exhaustively true, and the giant holds 99.91 % of the sector. The quasi-one-dimensional cells behave in the opposite way: at $(1, 1, 3)$ twelve non-singleton components share the sector, and at $(1, 1, 4)$ there are 66 components of size 2,496 and eight of size 42,240, none of them frozen and none the giant. Disconnection at those cells is an aspect-ratio effect, which is why the conjecture is stated for cubic cells only.

The first cubic cell is already out of reach, with roughly 2×10^{11} zero-flux states at $(2, 2, 2)$. The next anisotropic cell up, $(1, 2, 3)$, has 196,465,480 of them and is blocked three ways: its 96 bonds exceed the 64-bit state word that the enumerator and the component machinery both assume, the depth-first enumeration projects to about 8.5 hours at the measured leaf rate, and the flip graph would carry some 9.4×10^9 edges. The exact route stops here, so the evidence at cubic cells has to be of a different kind.

4 Random starts, and what a flip path certifies

The probe is simple. Draw a zero-flux state at random, and if it is not frozen try to build an explicit path of hexagon flips from it to a fixed reference state R . A path found certifies that the two states share a component. A path not found is inconclusive, and is never reported as a negative result.

The reference state is fixed by the convention `giant-start-1`. Starting from the Klein zero-flux state, which is frozen at every cell, the construction applies $4n_{\text{spins}}$ contractible directed-loop reversals at fixed $\mathbf{W} = \mathbf{o}$ to destroy the frozen tiling, then raises the flippable-hexagon count by simulated annealing using hexagon flips only, keeping the best of eight deterministic restarts. The annealing step is component-preserving, so the accessible sector D is defined as the component of R rather than assumed to be anything. At $(2, 2, 2)$ the resulting R has 40 of its 128 hexagons flippable.

Two samplers supply the starts, because neither is uniform and they fail in different directions. `sample_w0_dfs` runs a randomised ice-rule depth-first search to a single leaf and then drives the flux to zero with winding-changing loop reversals, giving a fresh state with no memory of any flip component. `sample_w0_loops` applies n contractible loop reversals to the Klein frozen state, preserving both the ice rule and $\mathbf{W} = \mathbf{o}$; with n small this manufactures near-frozen adversarial starts, which is what a hypothetical second non-frozen component should be most exposed to. Loop reversal is used only to manufacture starts and never in the measured dynamics, and the reason it is right here is the reason it is forbidden there: it can move between flip components.

The certificate itself is `certify_connected`, an annealed descent on the Hamming distance to R with the inverse temperature swept from 0.02 to 6 over 4×10^6 attempted moves and three restarts, and on failure a retry at 6.4×10^7 moves with six restarts. Every accepted move is a legal hexagon flip, checked against the alternation condition before it is applied, so a run reaching Hamming distance zero has walked a genuine path in the flip graph from the start state to R .

cell	family	starts	frozen	certified	uncert.	med. flip.
(2, 2, 2)	DFS	10,000	0	10,000	0	23
(2, 2, 2)	$n = 1$	1,000	11	989	0	14
(2, 2, 2)	$n \leq 16$	6,000	0	6,000	0	20-24
(3, 3, 3)	DFS	3,000	0	3,000	0	74
(3, 3, 3)	$n = 1$	1,000	1	999	0	16
(3, 3, 3)	$n = 2$	1,000	1	999	0	34
(3, 3, 3)	$n \leq 16$	5,000	0	5,000	0	47-77
total		27,000	13	26,987	0	

‘DFS’ is the fresh sampler and n is the number of loop reversals off the Klein state; the $n \leq 16$ rows aggregate $n \in \{2, 3, 4, 6, 8, 16\}$ and $n \in \{3, 4, 6, 8, 16\}$ respectively, of 1,000 starts each. Certified plus frozen exhausts every row, so no start was left uncertified anywhere. The last column is the median number of flippable hexagons, out of 128 at $L = 2$ and 432 at $L = 3$. The sweep shows the adversarial construction doing what it was built to do. One loop reversal off the Klein state leaves a median of 14 flippable hexagons

at $L = 2$ and 16 at $L = 3$, against equilibrium medians of 23 and 74. All thirteen frozen starts of the whole run came from $n = 1$ or $n = 2$, and every near-frozen start that was not itself frozen certified. Path lengths were not retained in the summary file, so nothing is claimed about how long the certifying walks were.

An earlier and smaller version of the same probe ran 780 zero-flux starts at cubic $L = 2$: 300 from the depth-first sampler and 120 each at $n = 2, 8, 64$ and 512 loops. None was frozen, none had two or fewer flippable hexagons, all 780 certified, and the median residual Hamming distance was zero. The 27,000-start run extends that by a factor of 22 at $L = 2$ and reaches $L = 3$, a cell of 432 spins, for the first time.

What makes the certificate worth quoting is that it was calibrated where the truth is known. At $(1, 2, 2)$ and $(1, 1, 4)$ the same procedure was run on 330 starts each and every verdict compared against the exact component decomposition. At $(1, 2, 2)$, 328 of 330 certified and all 328 were genuinely in R 's component. At $(1, 1, 4)$ only 17 of 330 certified; each of those 17 was genuinely in R 's component, and the remaining 313 genuinely were not. There were no false positives and no false negatives at either cell. The fresh-DFS group at $(1, 1, 4)$ certified 10 of 150, or 6.67 %, against the exact giant fraction of 6.74 % there. The annealed descent is therefore not vacuously permissive; it says no, correctly, 313 times out of 330 when the answer is no.

Neither sampler is uniform, and `sample_w0_dfs` rejects a fraction of its own attempts, which biases it further. The frozen fractions in the table are informative about the samplers and not about the uniform measure, and the 27,000 successes place no bound on the density of a hypothetical second non-frozen component.

5 Consequences for a mixing theorem

Since frozen states exist at every size, the spectral gap of the chain on the full zero-flux sector is exactly zero for all L , and a rapid-mixing theorem can only be stated on a component. That is a restriction on the statement, not a caveat attached to it. The project's chains start at R and move only by hexagon flips, so membership of $D = \text{comp}(R)$ holds by construction; what has to be measured is whether D is the giant.

It is not always. At $(1, 1, 4)$ the canonical reference state sits in a component of 42,240 states while the giant has 43,008, and the exact gaps on the two differ by a factor of 3.6. A

state drawn uniformly from the zero-flux sector at that cell is unreachable from the giant 93 % of the time.

If $(NF \Rightarrow G)$ holds, all of this collapses in cubic geometry. The accessible component is then the giant, characterising it reduces to enumerating the frozen states, and the frozen states are the part of the problem with structure: they are built by periodic tiling and by the Klein construction, they are periodic at $(2, 2, 2)$ and at $(3, 3, 3)$ with checked proofs, and their density falls with size, reaching 0.092 % already at $(1, 2, 2)$. A dynamical question about connectivity becomes a static question about a sub-extensive set. That is the reason to want the conjecture, and also the reason not to assume it.

6 What would refute it

One closed breadth-first search suffices. `icemix.component.bfs_component` starts from a single state held as a Python big integer, so it is not bound by the 64-bit word that limits the enumerators, and closes that state's component under hexagon flips subject to a state cap and a wall-clock cap. The asymmetry of the result is the point and is reflected in the return value. A search that closes is a proof, since the component then has exactly the states found, so a closed component containing a non-frozen state but not R is a counterexample; a search that hits either cap proves nothing and is reported as inconclusive. In the 27,000-start run the search was armed at a cap of 10^6 states and 1,800 seconds and never fired, because no non-frozen start went uncertified. It is tested against the exact decomposition at $(1, 1, 2)$, where it reproduces the 160-state component, and against the twelve frozen singletons at $(1, 1, 1)$.

The other refutation is an exact census at a cubic cell, which would settle the matter rather than merely refute it, and which is out of reach for the reasons given above. A cheaper intermediate target would be a census at a cell with two thick directions and one thin one beyond $(1, 2, 2)$; the nearest such cell, $(1, 2, 3)$, needs a new state representation before it needs anything else.

Two negative observations bound what the current evidence can be asked to do. The conjecture is false without the cubic restriction, as the $(1, 1, c)$ rows of the census show, so a proof has to use the geometry. And because neither sampler is uniform, a second non-frozen component could exist at cubic $L = 2$ or $L = 3$ provided it is missed by both,

which is a weaker thing to have ruled out than the raw count of starts suggests.

7 Related work

The graph-theoretic baseline is the cycle-reversal system. Reversing a directed cycle in an orientation preserves every in-degree, and conversely two orientations of a graph with the same in-degree sequence differ on a set of edges that decomposes into directed cycles, so reversing all directed cycles connects them; Gioan sets this inside the wider cycle-cocycle reversing system and counts its classes (Gioan 2007). Ice states are exactly the orientations with in-degree 2 everywhere, so the whole ice manifold is connected under reversal of arbitrary directed cycles, and it is the restriction of the cycles that creates the problem. On the torus, restricting to contractible cycles is what preserves the flux; restricting further to the shortest contractible cycles, the hexagons, is what admits frozen states. Read that way, $(NF \Rightarrow G)$ is a statement about how short the reversible cycles may be before connectivity fails, and frozen states are the exact locus of the failure. Counting Eulerian orientations, and the standard chain on them, go back to Mihail and Winkler (Mihail and Winkler 1996); in the planar case the chain that reverses directed faces is ergodic classically, and its mixing time on subgraphs of the triangular lattice is polynomial (Creed 2009). Connectivity is not the only obstacle even when it holds, as the six-vertex model on $\mathbb{Z}^2$ shows, where the local chain is torpidly mixing in the ordered phases (Liu 2018; Fahrbach and Randall 2019).

The dimer literature has met the same difficulty in the same dimension and has the sharpest results on it. For three-dimensional domino tilings the flip move alone does not connect a flux class; a second move, the trit, is needed, and the obstruction to flip-connectivity is a genuine integer invariant, the twist, rather than a pathology (Freire et al. 2022). Hartarsky, Lichev and Toninelli study which cycle lengths suffice for ergodicity of local dimer dynamics on hypercubic boxes and show that in three dimensions every configuration admits an alternating cycle of length at most six (Hartarsky, Lichev, and Toninelli 2024), the same shape of question as the one asked here and with the same answer, that short cycles are almost but not quite enough. Klivans and Saldanha survey the higher-dimensional picture and its open problems (Klivans and Saldanha 2025). Their general point, that beyond the planar bipartite case there is no canonical height function and ergodicity of the local dynamics stops being routine, applies directly to pyrochlore ice.

On the physics side the response to frozen and near-frozen configurations has been to replace the local move rather than to characterise where it fails. Melko, den Hertog and Gingras introduced loop moves for dipolar spin ice precisely because single-spin dynamics freezes below about 0.4 K in $\text{Dy}_2\text{Ti}_2\text{O}_7$, and the loop move, being ice-rule preserving, restores sampling of the quasi-degenerate manifold (Melko, Hertog, and Gingras 2001; Melko and Gingras 2004). The classical Monte Carlo community built the loop and worm algorithms with the same motivation (Barkema and Newman 1998; Shinaoka 2011; Otsuka 2014). Ergodicity breaking in this family is observed as well as simulated, since real-space imaging of vortex-frustrated artificial spin ice shows ergodicity transitions directly (Saccone et al. 2023). The question in this note is the complementary one: not how to repair the local chain, but whether the local chain is already ergodic off the set where it manifestly is not.

8 Artefacts

All paths are relative to the repository root.

path	contents
icemix/chain.py	build_flip_graph, components, and flippable_ref, the definition-level predicate the fast one is tested against
icemix/enumerate.py	enumerate_ice_states with zero_winding_only; the frontier program used for counts beyond enumeration
icemix/start.py	the giant-start-1 convention, the two samplers, certify_connected, verify_reference_in_giant
icemix/component.py	bfs_component, the bounded closure of one component at arbitrary spin count
icemix/_msc.c	msc_anneal_to_target, the annealed descent that applies only legal hexagon flips

path	contents
experiments/04_lorentzian_giant/	run_stage4abc.py, the 27,000-start run; results/part_4c.json, holding the second table, the recomputed (1, 2, 2) census and the run provenance
experiments/02_calibration/	results/results.json, holding the 780-start probe (connectivity_cubic_L2) and the certificate calibration (connectivity_calibration)
docs/reports/	2026-09-04-stage4abc.md, the run report with pre-flight, ledger and defects; 2026-08-11-stage1-falsifiers.md, the original census; 2026-09-04-certificates-sat.md, the certified frozen counts, including 612 at (2, 2, 2)

The census table was recomputed for this note by enumerating the zero-flux sector at each of the five cells, building the flip graph and taking the full multiset of component sizes. It takes about 28 seconds on one core for all five cells, and reproduces the earlier numbers exactly.

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